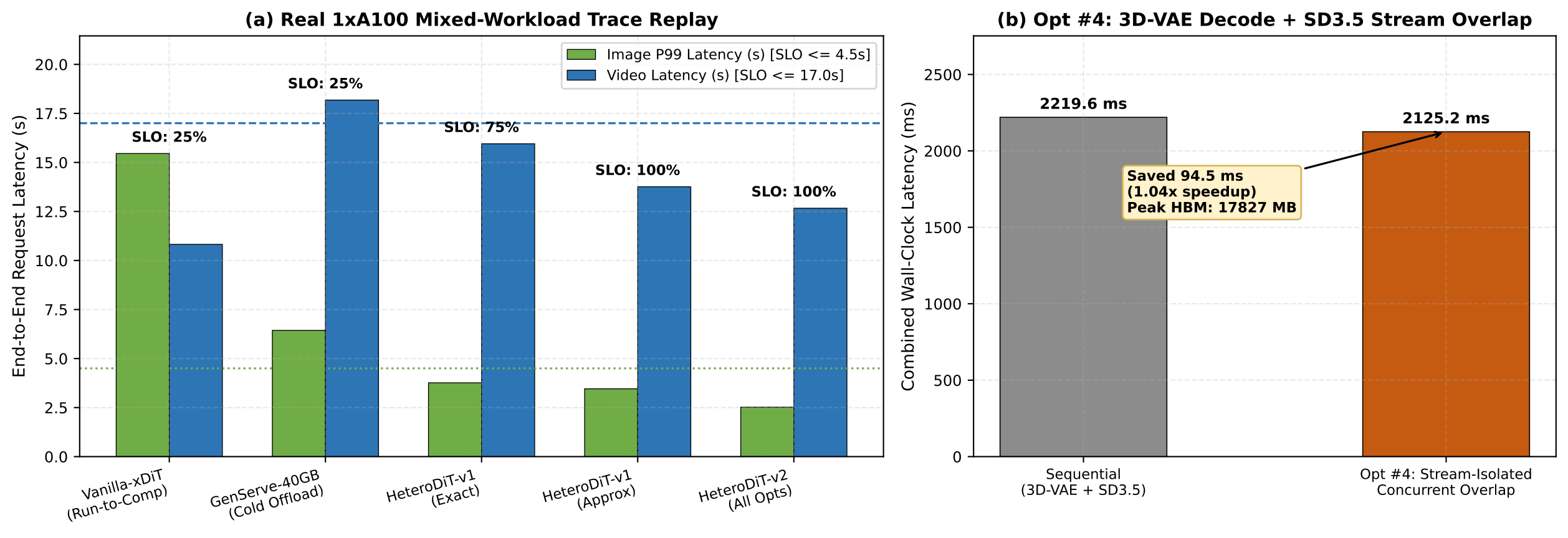
(a) Real 1xA100 Mixed-Workload Trace Replay
Image P99 Latency (s) [SLO <= 4.5s]
Video Latency (s) [SLO <= 17.0s]
End-to-End Request Latency (s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
SLO: 25%
SLO: 25%
SLO: 75%
SLO: 100%
SLO: 100%
Vanilla-xDiT (Run-to-Comp)
GenServe-40GB (Cold Offload)
HeteroDiT-v1 (Exact)
HeteroDiT-v1 (Approx)
HeteroDiT-v2 (All Opts)
(b) Opt #4: 3D-VAE Decode + SD3.5 Stream Overlap
Combined Wall-Clock Latency (ms)
0
500
1000
1500
2000
2500
2219.6 ms
2125.2 ms
Saved 94.5 ms
(1.04x speedup)
Peak HBM: 17827 MB
Sequential (3D-VAE + SD3.5)
Opt #4: Stream-Isolated Concurrent Overlap